



Design of thermal imaging-based health condition monitoring and early fault detection technique for porcelain insulators using Machine learning

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ABSTRACT

The inspection of insulator faults is an important task to prevent catastrophic failures in the operation of an electric substation. Manual inspection of overhead power line insulators can be very dangerous owing to the presence of high voltage in power sub-stations. Hence, in this paper, we present an infrared thermal (IRT) camera based non-invasive computer vision system for automatic monitoring and visual inspection of overhead on line power insulators. In the proposed work, initially, an optimal threshold method is applied to segment the region of interest (ROI) in IRT images. Subsequently, various geometrical, morphological, intensity and statistical features are computed from the segmented ROI, which are eventually utilized as an input to Gaussian kernel support vector machine to classify the different type of faults in insulator images. Computer vision based automatic inspection of insulators can play an important role from environment as well as human safety point of view. Timely inspection of insulators can ensure the environmental safety through prevention of fire that may be caused due to the insulator failures leading to the sudden breakdown of high-power lines. The proposed system achieved the true positive rate (TPR), and false negative rate (FNR) of 97.3%, and 2.66%, respectively. Whereas, the system obtained the Positive Predictive Value (PPV) and False Discovery Rates (FDR) of about 97% and 3%, respectively with the accuracy of 0.97 on the receiver operating characteristics (ROC) curve.

The performance of the proposed method is compared with other existing state of the art methods and found that our method outperformed over them. Hence, we recommend proposed system to detect the severity of faults in IRT images long before any catastrophic failures take place at power sub-stations.

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1. Introduction

Electrical Insulation is an essential component of all the electrical structures. The level of protection provided by the insulator substantially depends upon the magnitude of leakage current that pass through an insulator surface. The high leakage current can generate high voltage (i.e., 1000 V to 5000 V) at insulator dead end, which in turn depends upon environment conditions. Thus, the high intensity of leakage current might cause risk for human operators as well as for environment also. It may also lead to heavy power losses. The level of leakage currents depends on the health condition of insulator as the less faulty insulator leak out small magnitude of current, while critical faulty insulator release high leakage current. Therefore, computer vision-based insulator health condition monitoring system can help to determine the level of leakage current and can thereby protect the human life, as well as life of other creatures living in nearby natural habitats such as animals and birds. Electrical power substations play a vital role in catering the power requirements of domestic as well as industrial consumers (De Souza et al., 2017). The developing country like India has a vast network of power sub-stations for managing the transmission and distribution of power in remote as well as in urban areas (De Souza et al., 2017; Park et al., 2012). Power line insulators are the key components of high voltage transmission and power distribution systems, which are hanged on transmission towers or utility poles to occlude the flow of current between conductor and earth (Kind et al., 2016; Madi et al., 2016). The phenomenal increase of consumption of power in the recent times has led to a significant increase in the number of insulators in use on power lines. In general, Insulators are made up of various materials such as: glass, silicone, rubber, plastic, oil, wood, dry cotton, quartz, ceramic, etc. The porcelain/ceramic and toughened glass are the most common material, which are mainly used by electrical industries to construct the insulators (Ren et al., 2017). Glass insulators offers higher electrical resistant (between 500 kV/cm and 1000 kV/cm) and compressive strength (1.5 times) in comparison to that of ceramic insulators. However, the mechanical durability of ceramics is higher than those of glass insulators (Li et al., 2017; Shen-Pei et al., 2017). The environmental effects degrade the insulator performance leading to corona discharge and voltage flashover that may result in sudden breakdown of power grids or any catastrophe in high voltage transmission lines (Han et al., 2009). Some of the major causes of insulator failure includes: (i) Defective insulation material; (ii) Porosity in the insulation material; (iii) improper glazing on the insulator surface; (iv) flashover across insulator; (v) mechanical stresses on insulator. These defects enhance the flow of leakage current across its surface ultimately leading to a major catastrophic failure. Insulators being more prone to such failures needs regular manual inspection, which is in general carried out manually by human operators (Wang et al., 2015). The manual inspection of line insulators of overhead high voltage transmission lines is very risky, challenging and time consuming due to the involvement of high voltage, height of tower and other non-favorable environmental conditions encountered during the job assignment (Khalayli et al., 2013). Therefore, researchers are still striving hard to develop an efficient computer vision system for automatic fault detection and condition monitoring of various electrical equipment such as power transformers, composite insulators, high power electrical motors, and other rotating machinery etc. In recent years, Infrared thermography (IRT) has played crucial role for detection of possible faults in electrical equipment's in addition with ensuring the quality of other non-electrical products such as agricultural, manufacturing, and pharmaceuticals etc. (Jaffery and Dubey, 2014). The prime reason behind gaining the wide popularity of IRT technology for visual inspection of various high power electrical equipment is its non-contact and non-destructive features, which makes it capable to perform the fast and reliable inspection without interrupting the running operation of the equipment.

Further sections of the paper are organized as follows: A detailed description of approaches used for fault detection and health monitoring of insulators are given in Section 2. The proposed methodology is elaborated in Section 3. Sections 4 and 5 illustrate the implementation approach and simulation results, respectively. Finally, the work is concluded in Section 6.

2. Related work

The regular inspection and monitoring of insulators on overhead lines are difficult and very risky task due to the height of towers and location of power lines (mountains, terrains, and other non-accessible regions). To address the above issues, numerous methods (Shen-Pei et al., 2017; Han et al., 2009; Khalayli et al., 2013) have been suggested by eminent researchers and scientists in recent past years to inspect the insulator performance in laboratory experiments as well as in the outdoor fields, but none of methods have been proven useful and hence has not been adopted as a commercialized tool by electrical industries to perform the automated inspection and monitoring of overhead line insulators. In this section, we have briefly reviewed the various techniques and methods developed so far to monitor the health conditions of different types of electrical equipment's. However, out of the numerous techniques available in the literature, only few techniques focussed on non-invasive method for visual inspection and monitoring of porcelain insulators. This section also includes the advantages, limitations, and performance efficacy of the available methods.

For instance, Shen-Pei et al. (2017) used maximum entropy thresholding method and scale — invariant feature transform (SIFT) for fault diagnosis of insulators in infrared images. In this study, median filter was initially applied to denoise the insulator IR images, followed by elimination of background information using maximum entropy-based threshold algorithm. However, in the developed approach, it was difficult to precisely locate the affected regions in addition to higher computational cost and poor detection rate, owing to usage of complex feature matching algorithm and large variations in the faulty regions. Han et al. (2009) developed a novel electronic wave inspection system for the automatic inspection of porcelain insulators based on the analysis of low frequency even order harmonics of the basic

transmission line frequency (60 Hz) with the provision of mobile robot for keeping humans away from power transmission lines. The working principle of the developed approach is reliant upon the relationship between low frequency harmonic components and the defective characteristics of porcelain insulator. In this paper, authors used high order cumulants (HOCs) to carry out the retrieval of the low frequency harmonics from the complex noise. Similarly, Wang et al. (2015) developed a remote distance inspection system based on the laser ranging and image processing technology for the visual inspection of high voltage transmission lines with improved precision. The developed system involves the use of high-resolution camera, computer, two laser rangefinders, digital compass, and GPS to capture images and acquire information about the other parameters such as altitude, distance, and location of the object present under the transmission line. Khalayli et al. (2013) proposed a visual image sensor-based monitoring system for automatic detection and classification of STRI hydrophobicity associated with non-ceramic insulators. They extracted co-occurrence matrix-based features and utilized those as an input to develop polynomial classifier to perform the classification of insulator outer surface representing hydrophobicity. The proposed approach successfully achieved the classification rate of 97% for the stated problem. In year 2017, Pernebayeva et al. (2017) suggested a new technique based on Gabor and standard deviation filters for the visual inspection of ceramic insulators for identification of the water, snow, and ice particles on its surface. Using the proposed technique, authors successfully identified the snow and water droplets on the insulator surface with the recognition accuracy of 87%, but it could not be effectively utilized for detection of abnormal temperature distribution over the insulator surface being reliant upon only intensity and statistical features. In general, current discharging phenomenon through the insulator surface creates the abnormal temperature distribution on insulator surface, leading to the generation of a thermal image (Oberweger et al., 2014; Lv et al., 2017). Owing to the non-invasive capability, thermal imaging-based computer vision systems have gained a huge popularity and wider acceptance now days for the visual inspection and fault diagnosis of electrical equipment's (Bologna et al., 2002; Yassawi et al., 2015). For instance, Lv et al. (2017) developed a system based on image tracking technology for fault monitoring and visual inspection of insulators using infrared images. In their study, they employed histogram and Otsu threshold segmentation approach to calculate insulator's relative temperature with the aim of developing fault diagnosis system. However, the proposed system was tested only on 25 infrared images that limits its effectiveness for widespread use in practical applications. In Ref. Bologna et al. (2002), Bologna et al. conducted a series of test in High voltage laboratory at the University of Natal to demonstrate the capability of infrared and ultraviolet (UV) imaging techniques for the fault detection in composite insulators that occurs due to corona discharge and heating associated with leakage current flow and recommend the use of both techniques for the manual inspection of composite insulators in power sub-stations. However, they presented only manual techniques of using thermal vision and UV cameras for that task and did not propose any computer vision-based techniques to scan and inspect the insulators under test. Watershed city block transform algorithm was employed by Yassawi et al. (2015) to perform image segmentation for the fault recognition in IRT image. Water segmentation technique often encounter the over segmentation problem due to presence of noise and being sensitive to small intensity variations in image. Zhang et al. (2010) applied the morphological transform filter to denoise as well as improve the visual quality of airborne images for correcting the tilt angle of an insulator by applying Hough transform. Thereafter, texture features were computed to locate the insulator fault in airborne images. Infrared imaging is capable to delineate the abnormal temperature distribution on insulators caused due to the flow of surface leakage current and corona discharge effect more accurately and precisely in comparison to the manual examination from a safe distance (Rai et al., 2017). The remote visual inspection using thermal imaging system also ensure the human safety. Therefore, in this work, we present an infrared image-based fault detection and health condition monitoring system. The proposed work includes the development of a non-invasive thermal vision system for visual monitoring and classification of faults in respective thermal images of porcelain insulator to ensure their safety before any catastrophic failure takes place in electric sub-stations. Being non-invasive and highly sensitive for current discharging, thermal imaging-based vision system can monitor the high-power electrical equipment's from safe distance by detecting the discharging area very precisely (Sruthi and Sasikala, 2015; Meng and Application, 2011; Song et al., 2014). Poor health condition of an insulator allows the discharging current to pass through its outer surface, leading to proliferation of the abnormal temperature distribution over the surface (Ullah et al., 2017; Kuber and Dixit, 2014; Huang et al., 1979; Singh and Jaffery, 2018; Ding et al., 2018; Ledesma et al., 2018).

Our main contribution in this paper can be summarized as follows:

- (1) In this paper, we proposed an infrared imaging-based real time insulator health condition monitoring system. The proposed system is developed by integrating image enhancement, segmentation, and gaussian kernel-based support vector machine (GK-SVM) algorithm.
- (2) In this work, firstly an efficient pre-processing algorithm was implemented to enhance the quality of input image that was followed up by segmentation algorithm for the extraction of region of interest (ROI) from the segmented image.
- (3) In the second phase, we extracted the geometrical and statistical features and used them to build GK-SVM classifier to classify the insulator into less faulty, medium faulty and critical faulty categories.
- (4) The performance of the proposed system is compared with other existing state of the art method to evaluate the efficacy of our system.

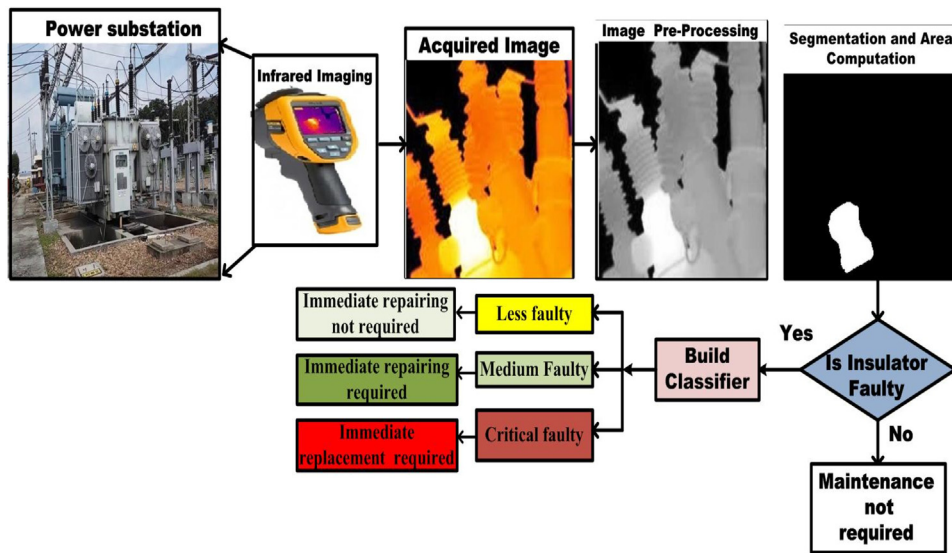


Fig. 1. Process flow diagram of proposed insulator health condition monitoring system.

Table 1
Fluke-Tis 20 thermal camera specifications.

S. No.	Parameters	Specification
1	Resolution	120 by 90 pixels
2	Display	3.5 Inch (landscape) 320 × 240 LCD
3	Temperature measurement range	−20 °C to 350 °C
4	Internal memory	4 GB
5	Built in digital camera	5 MP
6	Accuracy	±2 °C or 2%

3. Methododology

In this section, we present an efficient non-invasive real time visual inspection system. The proposed system comprises of the following main steps: (i) Image enhancement, (ii) region of interest segmentation and extraction, and (iii) development of GK-SVM based classifier. Fig. 1 shows block diagram of proposed system.

3.1. Non-Invasive Visual Inspection based Insulator Health Condition Monitoring (NVIHCM) system

The proposed NVIHCM system includes thermal image acquisition system (i.e., IR Thermography), pre-processing, optimal threshold-based image segmentation, feature extraction, and development of GK-SVM classifier. In this section, all the steps that comprises the proposed system, are discussed one by one in considerable detail as follows:

3.1.1. Image acquisition

Infrared thermography is a non-destructive temperature measuring technique that has gained wide acceptance for measuring the temperature variation over the surface of electrical gadgets for detection and prediction of possible faults in different electrical assets (Sruthi and Sasikala, 2015; Meng and Application, 2011; Song et al., 2014; Ullah et al., 2017). To develop NVIHCM system, a dataset comprising of 450 IR thermogram images was generated using a Fluke Tis-20 digital thermal camera. Table 1 provides the complete specifications of the Fluke Tis-20 digital thermal camera. The sample IR thermogram images acquired by this camera are shown in Fig. 2.

The acquired images had primely three types of faulty regions namely: critical faulty (CF), medium faulty (MF), and less faulty (LF). These three types of faults (i.e., CF, MF, and LF) were decided based on the geometrical and statistical features of the suspected faulty region. All the captured images were divided into training and testing samples into the ratio of 75 and 25 percent, respectively for developing the GK-SVM classifier (de Souza et al., 2014; Wei et al., 2018; Cuevas et al., 2011).

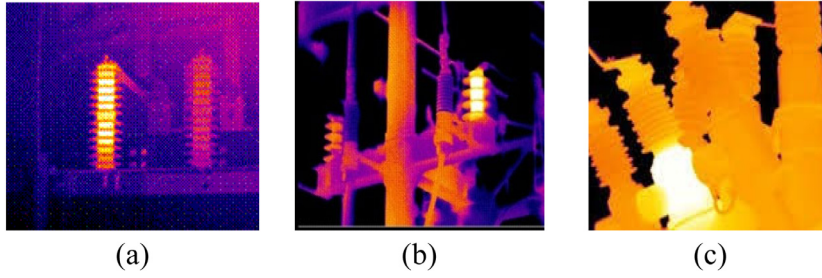


Fig. 2. Sample thermogram images of insulator captured by thermal camera.

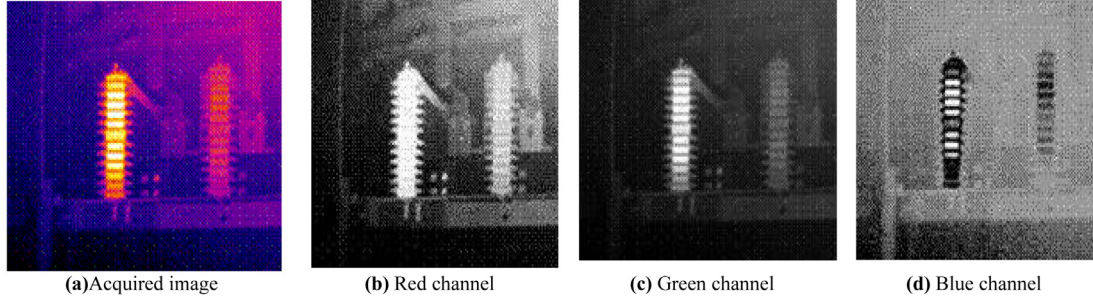


Fig. 3. (a-d). Sample of acquired images and their Red, Green, Blue channel.

3.1.2. Pre-processing

In general, thermogram images contain noise and other artifacts, which needs to be eliminated prior to segmentation to yield good detection and classification results. The pre-processing includes two major steps: (i) transformation of RGB images into intensity/ or Gray scale images; and (ii) image denoising. Both steps are essential and must be followed before applying any segmentation algorithm to improve the image quality of input thermograms.

RGB to Gray scale conversion

In general, the input thermogram images contains RGB pixel components that represents three channels viz., red, green, and blue. We transformed red, green, and blue channels into 2 dimensional single gray channel to reduce the time complexity, memory requirements, and to seek better visual representation of features in IRT images. Based on the channel analysis technique (Kumar and Verma, 2010), green channel provides more accurate information about the faulty candidates. Hence, only green channel was selected for subsequent processing to extract the object of interest. Sample thermogram and its corresponding red, green and blue channel after gray scale transformation is shown in Fig. 3(a-d).

Image De-noising

In general, IR thermograms contains noise and several other artifacts that must be eliminated prior to segmentation; otherwise, it may result in irrelevant object segmentation eventually leading to poor classification performance. The noise modifies the feature information in the thermograms by altering the intensity of image pixels. Therefore, image de-noising is considered as an important pre-processing step to make the thermogram suitable for subsequent processing and feature extraction. In this study, 2-dimensional median filter with the disk-shaped structuring element (SE) of size 5 by 5 pixels was used for noise reduction thereby improving the visual quality of thermograms (Jaffery et al., 2017). To determine the optimal size of SE to 5×5 , we conducted the pilot experiments on some sample images prior to the actual implementation of the segmentation algorithm and found it most suitable for the input images under study (Kuber and Dixit, 2014; Huang et al., 1979). The resulting images after applying median filter are shown in Fig. 4(a-f).

3.1.3. Image segmentation

In NVIIHCM, an optimal thresholding (Jaffery and Dubey, 2014) with smoothing technique is applied for segmentation of region of interest to explore the faulty regions. The mathematical expression of the binary image, which is obtained after applying the optimal threshold algorithm is given as:

$$B(i, j) = \begin{cases} 1; & \text{if } G(i, j) > \alpha \\ 0; & \text{if } G(i, j) \leq \alpha \end{cases} \quad (1)$$

where $G(i, j)$ and $B(i, j)$ represents gray scale image and transformed binary image, respectively and α is the specific threshold value needed to threshold the image. As already been stated in previous section that single channel gray scale

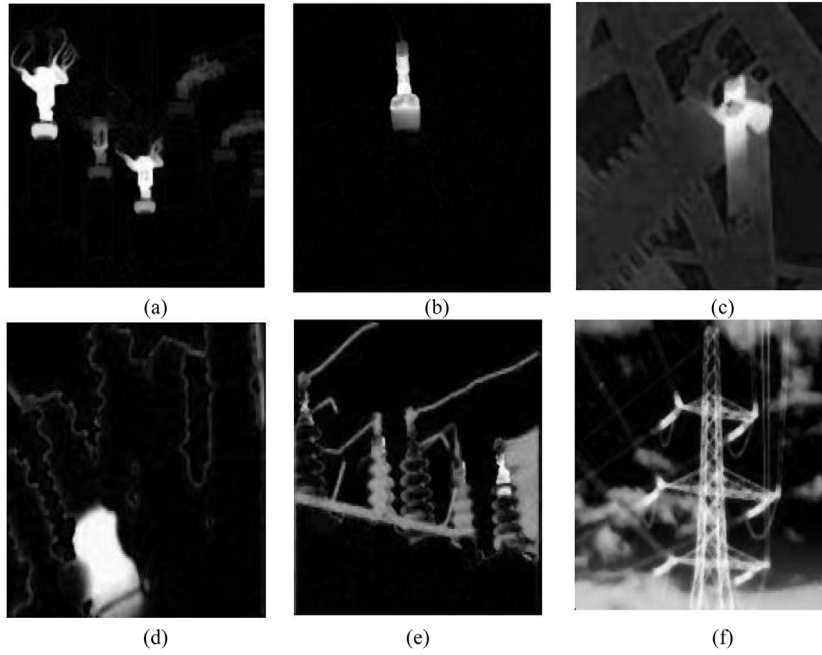


Fig. 4. (a–f). Sample of enhanced image after applying the median filter.

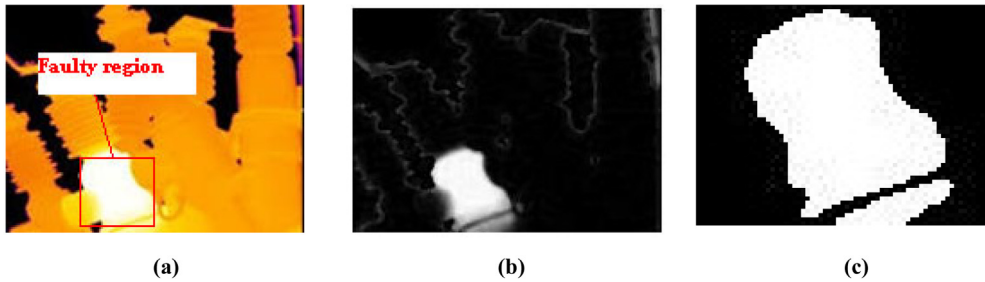


Fig. 5. (a) Original input thermogram, (b) image after RGB to gray conversion and median filter, (c) Segmented faulty region (enhanced view) obtained after applying threshold algorithm.

image was obtained after applying the gray scale transformation algorithm on the green channel color image. The gray scale image was segmented to yield the binary image as shown in (Fig. 5-c) using Eq. (1). Processing unit localizes the faulty region in the captured image and finally helps to extract the geometrical and statistical features from the segmented binary object (Singh and Jaffery, 2018; Mala and Sridevi, 2016).

3.1.4. Feature extraction

In general, features extraction and selection plays a critical role in building an efficient classifier and the features such as geometrical and statistical, which are listed in Table 2 are important indicator of size and shape of the suspected faulty regions. Based on the extracted features, the output class was labeled as any one of the three distinct classes viz., less faulty, medium faulty, and critical faulty. Less faulty image may be treated as a normal image as there are least chances of power failure owing to it. Hence, in this step, we focus on extraction of the geometrical and statistical features from the segmented image (Fig. 5-c). These features are listed in Table 2 and are described as follows:

Area: The area of a region is defined as the number of pixels in the region. It represents the category of the fault. Larger is the area of faulty region, more severe the fault is. It is mathematically expressed as:

$$A_r = \sum_{i=1}^n B(x, y) \quad (2)$$

where, A_r represents area of faulty region and $B(x, y)$ represents binary image.

Table 2
Details of extracted features.

Feature number	Features description	Acronyms/symbols
1	Area	A_r
2	Major axis	Mj_a
3	Minor axis (M_i)	Mn_a
4	Eccentricity, $Ecc(R)$	$Ecc(R)$
5	Perimeter	$P(R)$
6	Compactness	C
7	Mean Intensity#	μ_i
8	Standard deviation#	σ

Note- 1–5: Geometrical features, and 7–8: Intensity and statistical features.

Major axis: represents the longest chord of the segmented faulty region. The major axis of faulty region is shown in Fig. 6 and can be mathematically expressed as:

$$Mj_a = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \quad (3)$$

The major axis endpoints (x_1, y_1) and (x_2, y_2) are determined by computing distance between every border pixels in the object. The maximum distance between the border pixels is considered as major axis.

Minor axis: represents the shortest chord of the segmented faulty region. Mathematical expression of the minor axis is given by (4) as:

$$Mn_a = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \quad (4)$$

Eccentricity: It represent the ratio of the semi-major axis to semi-minor axis. It provides information about the shape of the segmented faulty region.

$$Ecc(R) = \frac{Mj_a}{Mn_a} \quad (5)$$

where, Mj_a is major axis feature of insulator faulty region and Mn_a is minor axis of training image region.

Perimeter

$$P(R) = \sum_i \sqrt{(x_i - x_{i+1})^2 + (y_i - y_{i+1})^2} \quad (6)$$

where, x_i and y_i are the coordinate of the i th pixel at the boundary region.

Compactness: It is expressed as the ratio of the square of the perimeter to the area of an segmented region that represents the fault. This is given as:

$$C = \text{perimeter}^2 / \text{area}$$

Solidity: represent the ratio of the area and convex hull area of the segmented region. Solidity of an object is calculated as follows:

$$S(R) = \frac{A_r}{CA_r} \quad (7)$$

where $S(R)$ depict the Solidity of region, and A_r and CA_r represent area and convex hull area of the segmented region, respectively.

Mean Intensity: mean intensity defines the average of pixels value of the segmented region and can be mathematically expressed as:

$$\mu_i = \frac{\sum_{i=1}^N x_i}{N} \quad (8)$$

Standard deviation: It express the range of the pixel values indicating the difference between the maximum temperature and minimum temperature of the segmented faulty region.

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu_{i+1})^2} \quad (9)$$

3.2. Building the GK-SVM classifier

Using the features extracted in the previous step, GK-SVM based classifier was built to take decision about the health condition of insulator. To build the classifier, statistical and geometrical features were initially turned into vector sets,

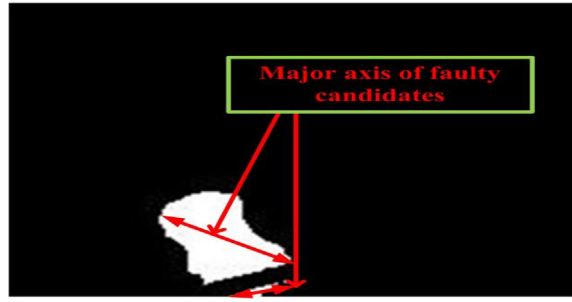


Fig. 6. Representation of major axis (geometrical feature) of faulty candidate.

which were subsequently divided into ratio of 70 and 30 percent for training and testing purpose. SVM selects support vectors from the input feature vector and draws a hyper plane to categorize the classes. The algorithm draw many hyperplanes with support features vectors but keep only those hyper planes, which shows maximum margin between the classes (de Souza et al., 2014; Wei et al., 2018). Mathematical expression of hyperplanes for two class decision making is shown using Eq. (10).

$$h(x_i) = \begin{cases} +1 & \text{if } f(x) \geq 0 \\ -1 & \text{if } f(x) < 0 \end{cases} \quad (10)$$

where, $h(x_i)$ represents the hyper plane of classifier and $f(x)$ represents the function of hyper plane. If the classifier classifies test point as +1, it indicates that the test point lies on the right side of hyperplane, hence the point belongs to the positive class. However, if the classifier classifies a test point as -1. It indicates that the point lies on the left side of the hyperplane, hence point belongs to negative class. This work developed non-invasive health monitoring system that classifies an insulator's faults into three different categories such as less faulty (class-0), medium faulty (class-1), and critical faulty (class-2). The performance of linear SVM is not suitable for multiclass problem. However, kernel based SVM obtains good results in case of multiclass problem. Therefore, in this work, we adopted gaussian kernel SVM (GK-SVM) for visual inspection and health condition monitoring of Insulators (Wei et al., 2018). Gaussian hyper plane function is mathematically expressed as:

$$f(x) = \sum_i^N \alpha_i y_i k(x_i, x) + b \quad (11)$$

where, α represents weight it may be 0, $k(x_i, x)$ represents the Gaussian kernel equation which is drawn with the help of support vector. Mathematical equation for Gaussian kernel is given by Eq. (12).

$$k(x_i, x) = \exp\left(-\frac{\|x - x_i\|^2}{2\sigma^2}\right) \quad (12)$$

4. Implementation approach

The process flow diagram of the proposed system is shown in Fig. 1. This process flow diagram includes acquisition of IR thermograms, image pre-processing, segmentation, feature extraction, building classifier, and controlling action. In this work, original insulator thermogram images were captured using Fluke Tis-20 thermal camera to develop an efficient insulator health condition monitoring system. The proposed methodology was implemented on MATLAB 2019 using hp AMD A6-7310 APU processor with 4 GB primary memory. Total 450 (150 images for each class) thermogram images were acquired and considered for the development of the proposed system. The Fig. 7(a) shows the sample images of insulators containing faulty regions. The faulty region depicted in Fig. 5(a) is annotated by red color square box. In order to make the input images more suitable for feature extraction, all the thermogram images were passed through a several pre-processing steps as discussed in detail in Section 2. The first step of pre-processing involves the decomposition of RGB color images into their three constituent color channels viz., red, green, and blue channels. Gray scale transformation algorithm was then applied on the individual color channels to get the better insight about visual features of the faulty regions in an insulator thermogram image. Green channel image was found more suitable for feature selection, when we conducted a visual examination on some pilot images. As we can notice in Fig. 3(c) that object of interest (i.e., faulty region) is more clearly visible in green channel image as compared to red channel and blue channel images, after gray scale transformation. Therefore, in this work, authors considered only the green channel images for transformation into gray scale images, followed by application of image thresholding algorithm to binarize the object of interest as illustrated in Fig. 5(c). However, an adaptive median filter was used to remove the small blemishes, and intrinsic noise. The image obtained after applying gray scale conversion and adaptive median filter is shown in Fig. 5(b). Here, we preferred to

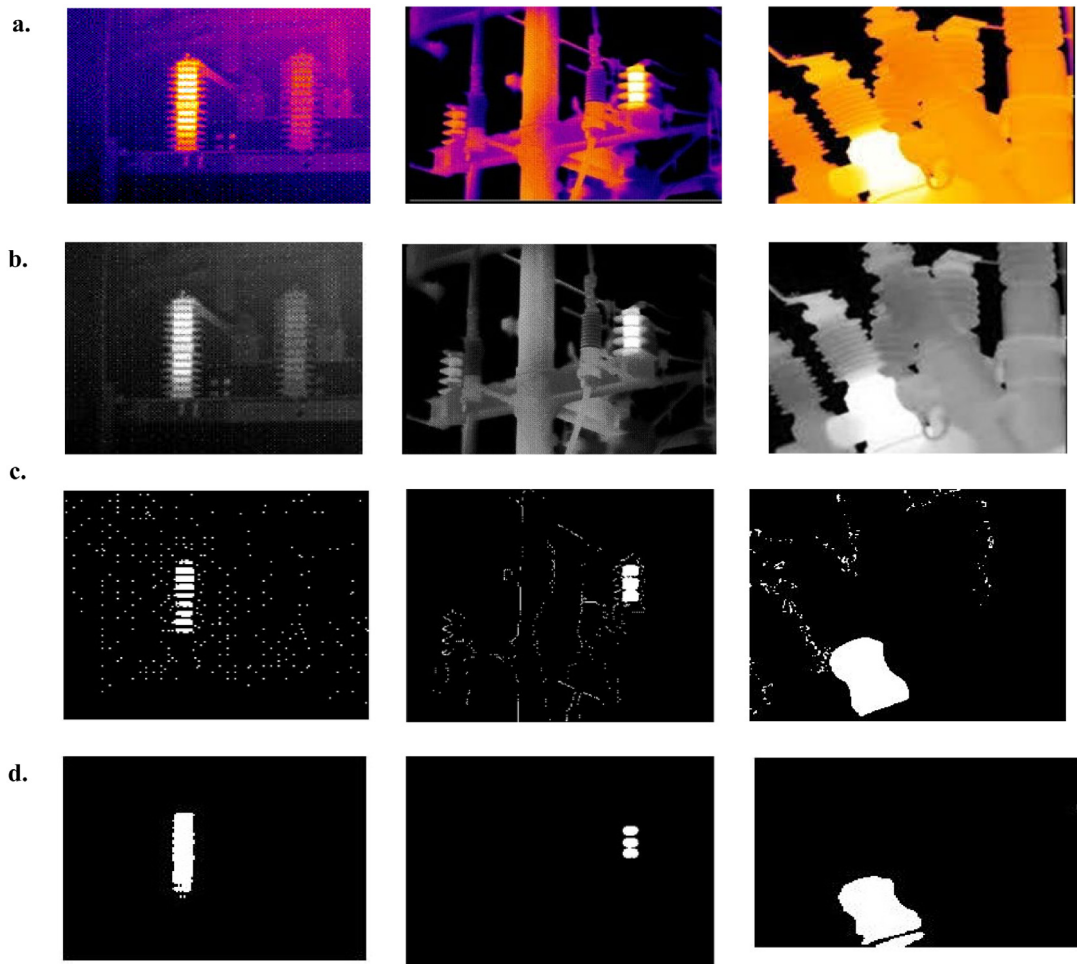


Fig. 7. (a) Original thermogram images, (b) Pre-processed image (Green channel image after RGB to gray scale conversion), (c) Segmented image using optimal threshold method, (d) Post processed image representing faulty regions (i.e. ROIs).

adopt adaptive median filter to smoothen the image due to its good response to supersede the noise without altering the important (i.e. especially the edge) features of ROI. Thereafter, we obtained the binary segmented image after applying the threshold algorithm on the resulting smoothened image. The enhanced view of segmented object is shown in Fig. 5(c). From the segmented object, number of statistical and geometrical features (as shown in Table 1) were extracted to be utilized as an input to build the Gaussian kernel based SVM classifier. Initially, GK-SVM classifier was trained using the geometrical and statistical features, which we computed from the segmented binary object (i.e., faulty region) and a corresponding target label was assigned. After the completion of the training phase of the classifier, its testing phase begins, wherein new features set which was derived from test sample images were fed as an input to the classifier. GK-SVM makes a comparison between the test feature set and training feature set to assign a class label to the test sample image.

5. Simulation results

This section presents the quantitative results of the proposed method. The proposed system is implemented on MATLAB 2019a software using personal computer with Intel core i7 (speed 3.7 GHz) processor and 8 GB RAM. A total of 450 sample images (i.e., 150 images of each class) were used for the performance evaluation of the proposed GK-SVM classifier. The images that were segmented after applying the segmentation algorithm were used as an input to the classifier as discussed in Section 3. The segmented images represent the faulty regions or region of interest (ROI). The dataset available in the form of ROIs was divided into the ratio of 75% and 25% for training and testing the classifier. Fig. 7 illustrates the results of various steps of the segmentation algorithm. Fig. 7 shows thermogram original image, image after pre-processing (i.e., conversion of green channel to gray scale image), image after threshold segmentation, and ROI. These steps have

already been explained in considerable detail in the above section. From the segmented image, the geometrical and intensity features such as area, major axis, minor axis, perimeter, eccentricity, mean intensity, and standard deviation were computed and used as an input for GK-SVM classifier. The geometrical features represent the shape and size of the faulty region indicating spread of leakage current over the region, while intensity features indicate the intensity of the leakage current. If the larger area of the porcelain insulator has been exposed to leakage current, in such case, the geometrical features will have the higher numerical values while the pixel value of the intensity feature shall depend upon the magnitude of the current flowing through the insulator surface.

All the features extracted from 75% of the total images were combined in a excel sheet to train the classifier. A separate excel sheet was also prepared to store the features of remaining images in order to test the classifier. The performance of the proposed classifier was measured based on True Positive Rate (TPR), Positive Predictive Value (PPV), False Discovery Rate (FDR), False Positive Rate (FPR), and accuracy. In recent studies, these metrics have been used quite successfully by various researchers for the performance evaluation of the classifiers (Cuevas et al., 2011; Kumar and Verma, 2010; Baly and Hajj, 2012). The following section provides the detailed discussion on these metrics.

5.1. Performance metrics

We used several metrics such as True Positive Rate (TPR), Positive Predictive Value (PPV), False Discovery Rate (FDR), False Positive Rate (FPR), and accuracy, to quantify and evaluate the performance of all the models under consideration. Over the years, these metrics have been widely used in the literature to judge the performance of classifiers (Cuevas et al., 2011; Kumar and Verma, 2010; Baly and Hajj, 2012). These metrics can be defined as follows:

True Positive Rate (TPR)

The true positive rate is also known as **sensitivity**, is used to measure the percentage of actual positive candidates, which are correctly identified by the system. The TPR can be mathematically expressed as:

$$TPR\% = \frac{TP}{TP + FN} \times 100$$

In general, the system with higher value of TPR (or sensitivity) and lower false positive rate claims to have higher efficacy (Singh and Zaffery, 2017).

Positive Predictive Value (PPV)

The positive predictive value measures the probability of an abnormal region being predicted as a abnormal region by the system. The **Positive Predictive Value is defined as**

$$PPV\% = \frac{TP}{TP + FP} \times 100$$

where a “True positive (TP)” is the event that the test makes a positive prediction, and the subject has a positive result under the gold standard, and a “False positive (FP)” is the event that the test makes a positive prediction, and the subject has a negative result under the gold standard. The ideal value of the PPV, with a perfect test, is 1 (100%), and the worst possible value is zero.

False Discovery Rate (FDR)

The false discovery gives information about how many normal (false) regions have been detected by the system as the abnormal regions among the overall detected samples. FDR can be mathematically denoted as:

$$FDR\% = \frac{FP}{TP + FP} \times 100$$

False-positive rate (FPR)

False positive rate (FPR) is also referred as fall-out or false alarm ratio. FPR is calculated as the ratio of the number of negative events wrongly categorized as positive (false positives) and the total number of actual negative events (regardless of classification) as expressed as follows: .

$$FPR\% = \frac{FP}{FP + TN} \times 100$$

Accuracy (ACC)

Accuracy is simply calculated as a ratio of correctly predicted observation to the total observations. Practically, accuracy score of any model should be as high as possible and it can be mathematically defined as:

$$ACC\% = \frac{TP + TN}{TP + TN + FP + FN} \times 100$$

True Positives (TP): The system predicted faulty insulator as faulty is known as True positive prediction.

True Negatives (TN): The system predicted non-faulty insulator as non-faulty is known as True negative predictions.

False Positives (FP): The system predicted non-faulty insulator as faulty is known as False positives.

False Negatives (FN): The system predicted faulty insulator as non-faulty insulator is known as a False-negative prediction.

Table 3
Performance parameters of different type of SVM prediction model.

SVM type	Accuracy	Prediction speed	Training time
Linear	96.4%	4900 ob/s	11.614 s
Quadratic	96.8%	4300 obs/s	15.012 s
Cubic	85.9%	6500 obs/s	76.967 s
Fine Gaussian (proposed GK-SVM method)	97.0%	5800 obs/s	10.134 s
Medium Gaussian	94.5%	2900 obs/s	12.938 s
Coarse Gaussian	85.0%	5100 obs/s	14.309 s

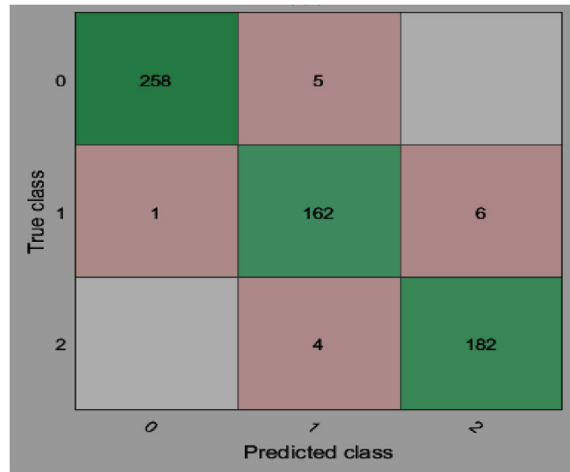


Fig. 8. Prediction matrix of proposed system.

5.2. Performance comparison

In this section, we present the simulation results of different variants of SVM classifiers and compared their performance with those of other competitive machine learning and deep learning models.

Different SVM model were trained one by one and observed their performance. According to those performances, this work has chosen the better system. Performance parameters detail such as prediction speed (observation per second), required training time (Sec) and percentage accuracy of different SVM model is shown in Table 3.

It is clear from Table 3 that fine Gaussian kernel based SVM predictor gives better accuracy, prediction speed and requires less training time. Therefore, in this study, we adopted Gaussian kernel based SVM model to develop insulator health monitoring system. The performance measures of the developed system are computed based on confusion matrix in terms of TPR, FNR, PPV, FDR, and accuracy. The confusion matrix of all the classification models is illustrated in Fig. 8. Horizontal axis of the confusion matrix shows the predicted label and vertical axis of confusion matrix shows the true label of each class.

Confusion matrix presents the prediction score of all the classification models including the proposed system. In confusion matrix, class 0, class 1, and class 2 represents the less faulty medium faulty, and critical faulty regions, respectively. From the confusion matrix, the proposed system predicted 258 faulty regions (FRs) successfully out of 263 FRs for class 0 and predicted 5 FRs as class 1. Similarly for class 1, the system predicted 162 FRs out of 169 FRs successfully, followed by making 1 prediction for class 0 and 6 predictions for class 2. Whereas proposed system predicted 182 faults out of 186 faults as class-2 and remaining 4 faults as class-1 successfully. Table 4 illustrate the summary of the information given in Confusion matrix.

In Table 4, T (0), T (1) and T (2) represent the true value of each class, while P (0), P (1) and P (2) represent the predicted value of each class. Table 5 presents the value of TPR, and FNR, which are calculated using the confusion matrix. From the results, it can be noticed that proposed system achieved the TPR and FNR value of 98%, 96%, 98% for class 0, 1, and 2, respectively, while system attained the FNR value of 2%, 4%, and 2%, respectively, for these classes. PPV and FDR have also been computed for the proposed system as illustrated in Table 6. Table 6 concludes that system obtained the PP value >99% for class 0, 95% and 97% for class 1, and class 2, respectively. Whereas the values of FDR were determined as 3%, 5%, and less than 1%, for class-2, class-1, and class-0, respectively.

ROC curve of proposed system for class-0, class-1 and class-2 is shown in Fig. 9. ROC curve for each class is plotted and analyzed based on accuracy point and system got remarks accordingly. Summary of accuracy point for each class is shown in Table 7 and a remark is also given about the system according to their accuracy point. According to accuracy point of the system it is concluded that proposed system monitors the health excellently.

Table 4

Summary of the observational (Prediction) data of confusion matrix.

Combination of true and predicted class	Observational data
T (0) P (0)	258
T (0) P (1)	5
T (0) P (2)	0
T (1) P (0)	1
T (1) P (1)	162
T (1) P (2)	6
T (2) P (0)	0
T (2) P (1)	4
T (2) P (2)	182

Table 5

Prediction summary of proposed system in terms of True positive and False negative rate.

	Class-0	Class-1	Class-2	Overall performance
TPR	98%	96%	98%	97.3%
FNR	2%	4%	2%	2.66%

Table 6

Prediction summary of proposed system in terms of PPV and FDR.

	Class-0	Class-1	Class-2	Overall performance
PPV	>99%	95%	97%	>97%
FDR	<1%	5%	3%	<3%

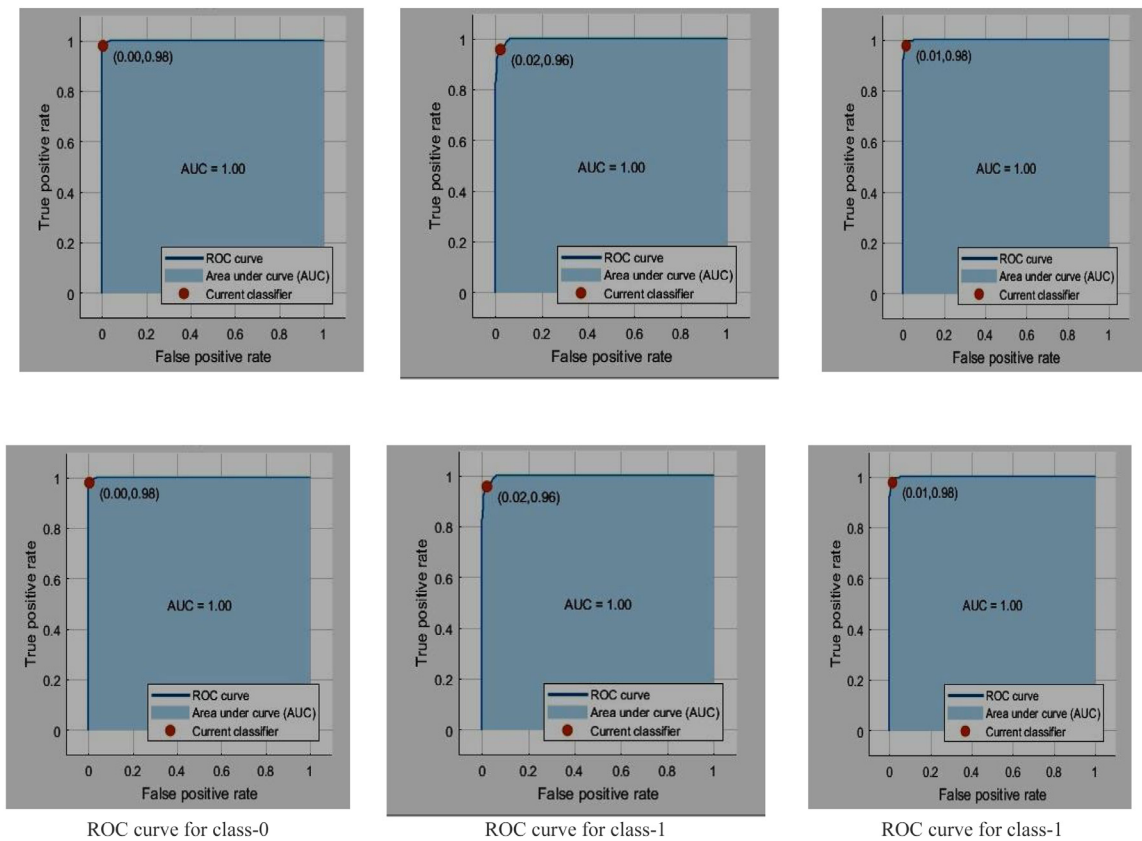
**Fig. 9.** ROC curve for class 0, 1, and 2.

Table 7
Accuracy point in ROC and remarks according to accuracy point.

Positive class	Accuracy point	Prediction remarks
Class-0	(0.00,0.98)	Excellent
Class-1	(0.02,0.96)	Excellent
Class-2	(0.01,0.98)	Excellent
Overall performance	(0.01,0.97)	Excellent

Table 8
Quantitative comparison of proposed method with deep learning-based approaches.

Model name	Accuracy	Average processing time	GPU required
Resnet-50 (He et al., 2016)	92%	4900 ob/s	Yes
Deep CNN (Han et al., 2020)	89%	4700 ob/s	Yes
Proposed method	97%	5800 obs/s	No

5.2.1. Comparison of proposed road detection system with current state of art methods

Currently, CNN and deep learning based methods are considered to be one of the most popular methods for developing vision-based detection systems. In CNN and other deep learning based methods, the performance of the system depends on accuracy, volume of data and processing time. Hence, we compared the performance of the proposed system with the CNN and other deep learning based approaches in terms of accuracy, processing time and GPU requirements, which is shown in Table 8. Results shown in Table 8 indicates that the proposed system was able to monitor the health condition of insulator with 97% accuracy and average processing time of only 0.58 s with no requirement of GPU. The proposed system is faster as compared to the deep learning based method. In addition, it does not require costly hardware, and is a noninvasive system. Therefore, it can be easily adopted for aforementioned application at wider scale due to its low cost, faster response and high accuracy.

6. Conclusion

This work proposed a non-invasive technique for the fault detection and health condition monitoring of insulators in power sub-stations. Thermal imaging technique is employed to capture the occurred fault in insulator images and eventually developed the insulator health condition monitoring system using image processing techniques and Gaussian Kernel based SVM classifier model. We adopted optimal threshold algorithm to partition an insulator image into faulty and non faulty regions and thereafter extracted morphological and statistical features from the faulty regions in order to build the GK-SVM classifier model for determining output response in terms of class labels assigned to the faulty regions. The developed model is trained to form the fault classification model. Detection and health monitoring performance of the model is also analyzed to check the reliability of the system. True positive rate of the system is 97.3%, false negative rate of the system is 2.66%, Positive Predictive Value is greater than 97% and False Discovery Rates of the system is less than 3%. Accuracy point of the system on ROC curve is 0.97, which indicates that the system classified insulator faults excellently.

CRedit authorship contribution statement

Laxman Singh: Implemented the proposed algorithm, Wrote the first draft of the manuscript. **Altam Alam:** Material preparation, Data collection, Analysis and carried out the implementation of GK-SVM classifier. **K. Vijay Kumar:** Material preparation, Data collection, Analysis. **Devendra Kumar:** Commented on the manuscript to improve the quality of the paper and thus helped in writing the final draft of the manuscript. **Parvendra Kumar:** Commented on the manuscript to improve the quality of the paper and thus helped in writing the final draft of the manuscript. **Zainul Abidin Jaffery:** Gave suggestions to improve the efficacy of the proposed model, and commented on the manuscript to improve the quality of the paper.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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